The walkthrough, run in Excel

threshold-engine on public data, 13 October 2026. Excel 16.0, through its automation interface.

433 of 433 cells as expected, in 30 of 30 steps.

What was done

The expected answers come from `expected.py`, from the source files and `scale.py`. 355 of them were written and committed before Excel was first opened (commit `6f3505d`); the rest were added after a run, each listed below under What I got wrong. `walk.ps1` then built the workbooks with the same command a person types, opened them in Excel, typed each step's values into Assess's inputs, let Excel recalculate, and read every cell back from Excel. Nothing was read from Python. `compare.py` put each reading beside its expected answer: a number must agree to within 1e-9, text exactly. After every step the typed cells were put back, and the Check column read blank again.

The card lines with the crisis and the pandemic left out (step W5) are held to a record made a week earlier, not to `scale.py`: LOG.md, 7 October 2026, **3.171 / 3.464 / 4.050 / 5.981**, with 25 / 22 / 29 / 55 / 4 quarters kept in each score. One card line is tied out separately to the Federal Reserve's own table (`docs/tie-out/cards-moderate-line-2026-10-13/`).

Result, step by step

Step	What	Cells	Result
W1	Build the four-product workbook and open it	9	as expected
W2	Thresholds as built: every product, nothing left out	48	as expected
W3.1	Assess: pick Credit card	17	as expected
W3.2	Assess: pick Mortgage	17	as expected
W3.3	Assess: pick Home equity	17	as expected
W3.4	Assess: pick Other consumer	17	as expected
W4	Leave out the financial crisis on Credit card	22	as expected
W5	Add the second leave-out: the pandemic (the rehearsal settings)	41	as expected
W6	Moderate's half-width 0.5 to 0.75 (H6)	31	as expected
W7	Moderate-Low step 1.0 to 1.5 (I6)	31	as expected
W8	High's z 2.5 to 3.0 (J6)	31	as expected
W9a	On the line takes the worse score (half-width 0)	22	as expected
W9b	On the line takes the better score (half-width 0)	22	as expected
W10.1	Refused: a leave-out with a start and nothing else	4	as expected
W10.2	Refused: a second leave-out in reverse order	4	as expected
W10.3	Refused: a leave-out with no reason	4	as expected
W10.4	Refused: a half-width below 0	3	as expected
W10.5	Refused: a z of 0	3	as expected

W10.6	Refused: a step typed as text	3	as expected
W10.7	Refused: too little history kept	3	as expected
W10.8	Refused: the median at or below zero (Mortgage, keeping 2011Q4 to 2026Q2)	3	as expected
W10.9	Refused: Moderate-Low inside Moderate	3	as expected
W10.10	Refused: High inside Moderate-High (z 0.3)	3	as expected
W10.11	Refused: High beyond the worst quarter kept (z 8)	3	as expected
W10.12	Refused: Moderate-Low at or below zero (step 6)	3	as expected
W11	Run tab: each file's fingerprint and the decisions as typed	20	as expected
W12	Chart: switch Mortgage off	2	as expected
W13	Evidence and Backtest for Credit card	11	as expected
W14a	A lower-is-worse score, on the line takes the worse score	18	as expected
W14b	A lower-is-worse score, on the line takes the better score	18	as expected

What it looked like

Net charge-off rate (TTM) — candidate thresholds

Live: recalculates from your choices on the Assess tab. Candidates for the bank to accept, adjust or reject.

Product	Median kept	Typical yearly move	Worst kept	2 Moderate-Low from	3 Moderate from	4 Moderate-High from	5 High from	Latest quarter	Latest (%)	Score	Rating	Check
Credit card	3.978	0.767	10.425	3.211	3.595	4.361	6.684	2026Q2	3.978	3	Moderate	
Mortgage	0.122	0.058	2.490	0.064	0.093	0.151	0.500	2026Q2	0.000	1	Low	
Home equity	0.182	0.095	3.140	0.087	0.135	0.230	0.685	2026Q2	-0.019	1	Low	
Other consumer	1.288	0.293	3.335	0.995	1.141	1.434	2.222	2026Q2	1.776	4	Moderate-High	

Scores: 1 Low, 2 Moderate-Low, 3 Moderate, 4 Moderate-High, 5 High. A loss at or below zero is always Low. Every term is defined on the Glossary tab.

W1–W2. Thresholds as built: every product, Check blank.

Periods left out: pick the product, the first and last quarter, and say why

Product	From	To	Reason	Status

Left-out quarters stay on the chart as grey dots over a dark strip; the lines are set from the rest. Clear a row (all four cells) to put its quarters back.

Your settings: one row per product (the product picked above is highlighted)

Product	Width of Moderate	Moderate-Low begins	How rare High is (z)	A value on a line takes the
Credit card	0.5	1	2.5	worse
Mortgage	0.5	1	2.5	worse
Home equity	0.5	1	2.5	worse
Other consumer	0.5	1	2.5	worse

W1. Assess's inputs as built, under its chart: Periods left out and Your settings.

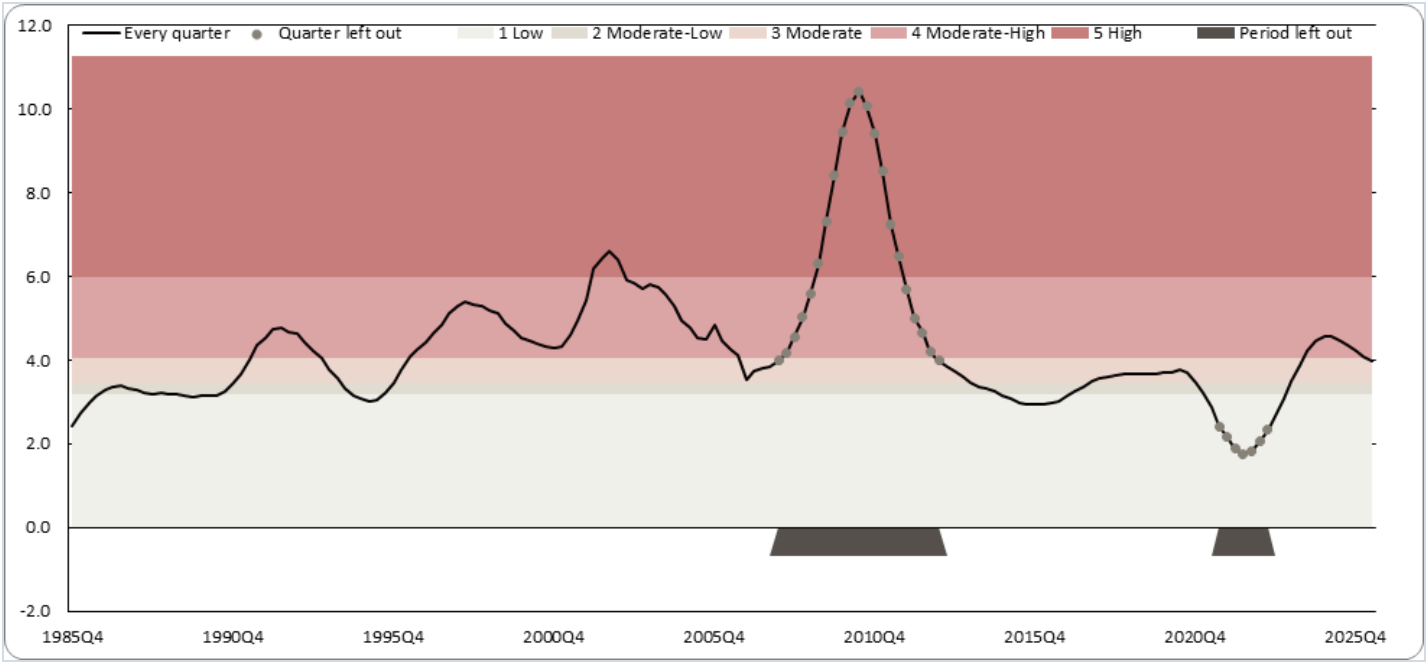
Net charge-off rate (TTM) — candidate thresholds

Live: recalculates from your choices on the Assess tab. Candidates for the bank to accept, adjust or reject.

Product	Median kept	Typical yearly move	Worst kept	2 Moderate-Low from	3 Moderate from	4 Moderate-High from	5 High from	Latest quarter	Latest (%)	Score	Rating	Check
Credit card	3.757	0.586	6.603	3.171	3.464	4.050	5.981	2026Q2	3.978	3	Moderate	
Mortgage	0.122	0.058	2.490	0.064	0.093	0.151	0.500	2026Q2	0.000	1	Low	
Home equity	0.182	0.095	3.140	0.087	0.135	0.230	0.685	2026Q2	-0.019	1	Low	
Other consumer	1.288	0.293	3.335	0.995	1.141	1.434	2.222	2026Q2	1.776	4	Moderate-High	

Scores: 1 Low, 2 Moderate-Low, 3 Moderate, 4 Moderate-High, 5 High. A loss at or below zero is always Low. Every term is defined on the Glossary tab.

W5. The rehearsal settings for cards: 3.171 / 3.464 / 4.050 / 5.981, as recorded on 7 Oct.



W5. Assess, cards: five bands, the two periods left out as grey dots and dark strips.

Assess and set one product's thresholds

Pick a product. The bands are its five scores and the line is its history. Leave periods out and change your settings bel

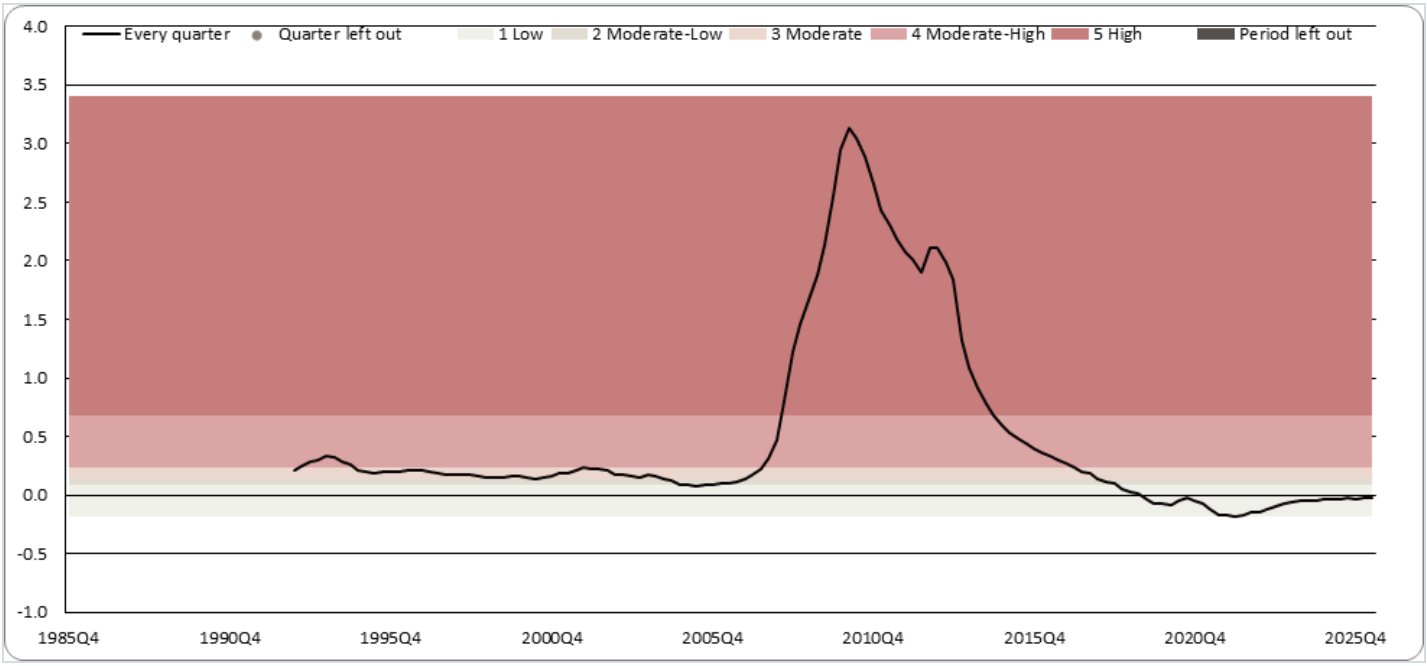
Product Credit card **Latest** **2026Q2: 3.978 -> 3 Moderate**

Check

Score	Rating	From	To	Quarters kept	Share kept	Every quarter	Share of every quarter
1	Low		3.171	25	19%	32	20%
2	Moderate-Low	3.171	3.464	22	16%	22	13%
3	Moderate	3.464	4.050	29	21%	31	19%
4	Moderate-High	4.050	5.981	55	41%	63	39%
5	High	5.981		4	3%	15	9%

Counts use the same rule as the score: a value on a line takes the score Your settings says. A loss at or below zero falls in score 1.

W5. Assess, the quarters in each score: 25 / 22 / 29 / 55 / 4.



W3.3. Assess, home equity, after the fix: every band, Low reaching below zero.

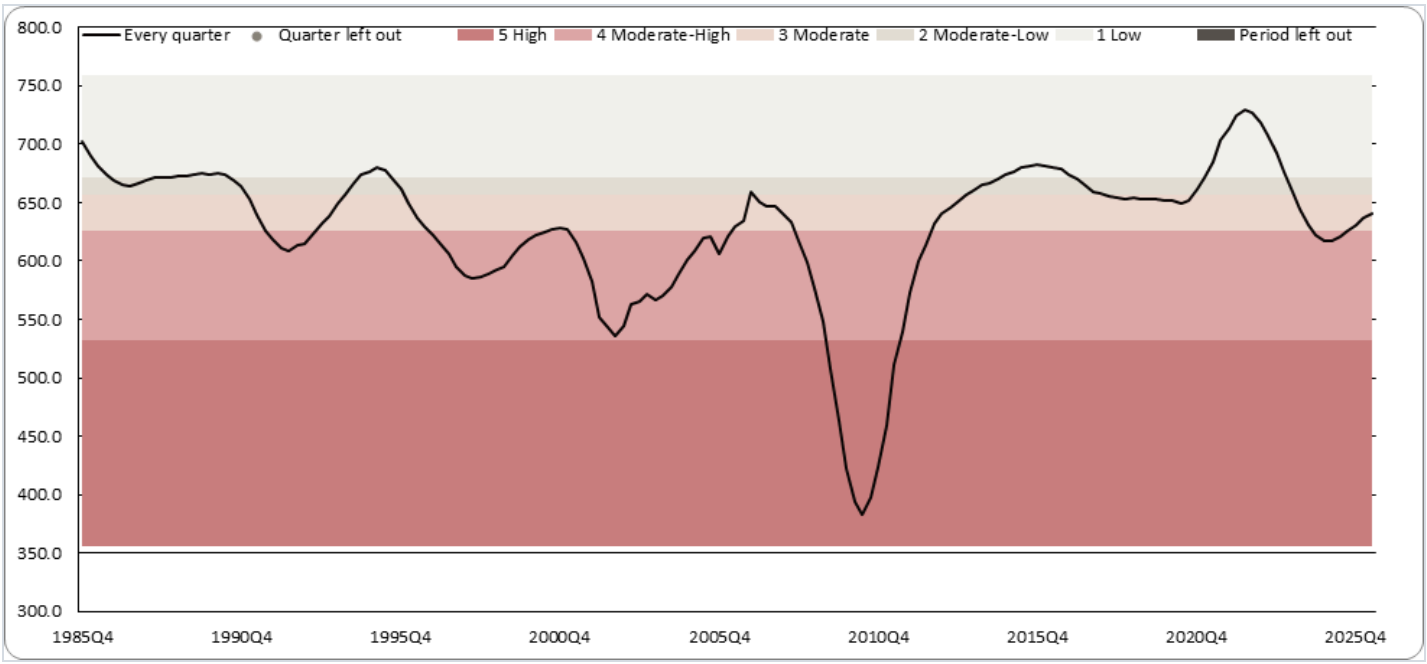
Score — candidate thresholds

Live: recalculates from your choices on the Assess tab. Candidates for the bank to accept, adjust or reject.

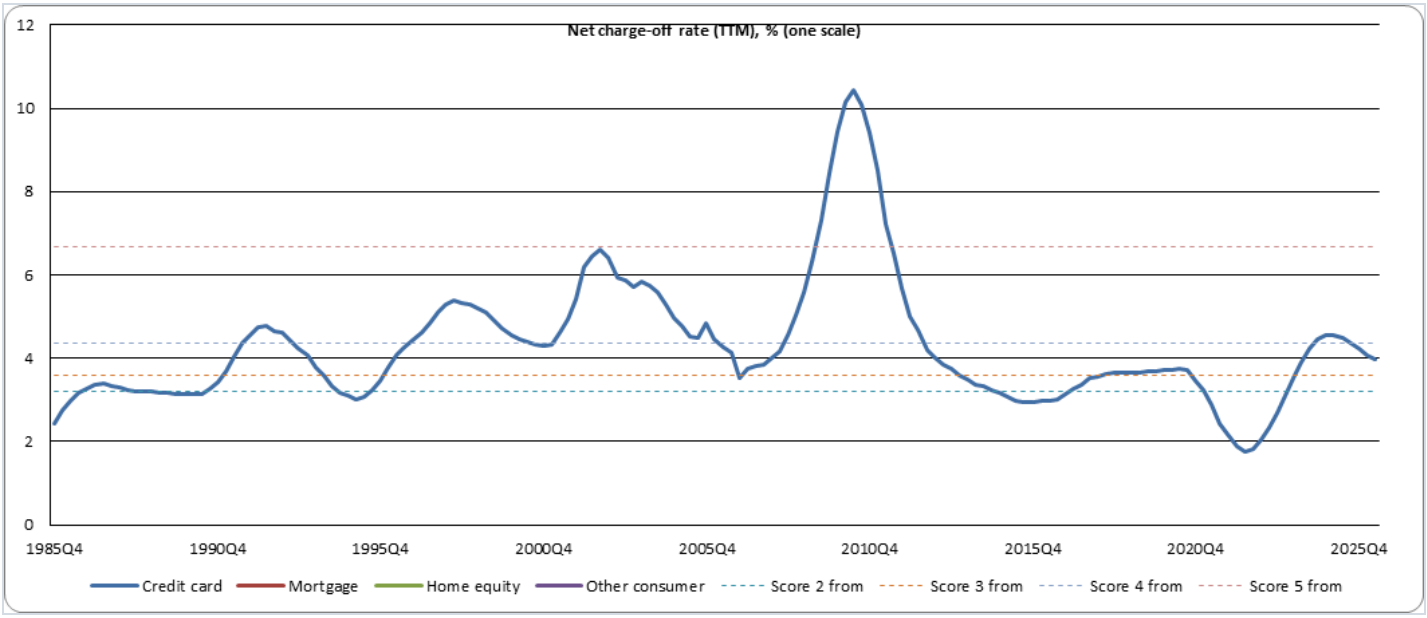
Product	Median kept	Typical yearly move	Worst kept	2 Moderate-Low from	3 Moderate from	4 Moderate-High from	5 High from	Latest quarter	Latest (points)	Score	Rating	Check
Score	640.880	30.660	383.000	671.540	656.210	625.550	532.650	2026Q2	640.880	3	Moderate	

Scores: 1 Low, 2 Moderate-Low, 3 Moderate, 4 Moderate-High, 5 High. Every term is defined on the Glossary tab.

W14. A lower-is-worse score: the lines fall from 2 to 5.



W14. Its Assess chart, after the fix: the axis starts just under the data, not at zero.



W12. Chart with Mortgage switched off; the legend below the plot, after the fix.

threshold-engine	0.11.0				
Built	2026-10-13T17:32:20				
Measure	Net charge-off rate (TTM)	%	higher_is_worse	smoothing 1	floor at ze
Scale, starting values	Moderate half-width 0.5	Moderate-Low step 1	High z 2.5	on the line: worse	
Backtest half-lives	none				
Backtest settings	percentiles 50/75/90/95	horizon 4	min history 41		
Product	File	SHA-256	Rows	First	Last
Credit card	tests\data\cards_nco_ttm.csv	8291d7d2b4ef971230fcf7271d2e49ed4b16a52f65612b1512d094cb3c69b1d0	163	1985-10-01	2026-04-0
Mortgage	tests\data\mortgage_nco_ttm.csv	fc768d8777b5e4c553c080fe590559c058a6de1cb6e127d90d56172b9bfedbb9	139	1991-10-01	2026-04-0
Home equity	tests\data\home_equity_nco_ttm.csv	393d5193027113ebb2383ea7e9be86cb8426ef69625acffd6012bef6a9557e09	135	1992-12-31	2026-06-3
Other consumer	tests\data\other_consumer_as_filed_nco_ttm.csv	0a46094231be4b0829b260adcf354cc89d8e7a37cf7e505fb2e75e9b7b52e6ae	135	1992-12-31	2026-06-3
Periods left out, as on Assess (live Product		From	To	Reason	Status

W11. Run: each file's SHA-256, equal to the files' own.

What it found

- **Assess drew the wrong bands for three of the four products.** For a product whose data starts after the first quarter on the calendar (mortgage 1991, home equity and other consumer 1992), the chart's top and floor were `MAX` and `MIN` over a column holding `#N/A`, so both were `#N/A`: Low and High vanished and the other bands were drawn from zero. Every count was right; only the picture was wrong. The tests charted cards only, which covers the whole calendar. Fixed with a helper column that is blank, not `#N/A`, and a test for each product.
- **A credit score drew from zero.** For a measure not floored at zero the chart's floor now sits just under the data, and the axis, which Excel starts at zero unless told otherwise, is given a fixed round minimum at build. The strip for periods left out now stands on an invisible base so it no longer pulls the axis down.
- **The Chart tab's legend covered the cutoff lines'** right-hand ends. It sits below the plot now.
- **WHAT-IT-DOES.md said the Glossary has 33 terms; it has 35.** The script's expected answer came from that document, and Excel disagreed. The document was wrong and is corrected.
- **The card file in the repository is rounded to three decimals**, which moves every card line by 0.0005 against the Federal Reserve's own figures and changes no score: see the tie-out.

What I got wrong

- The script's first card reference was LOG.md of 6 October, which belongs to the scale retired that evening. The 7 October record is the one for this scale; the script stopped on the mismatch before anything was run.
- Four readings failed on the first run because of the runner, not the workbook: a formula that returns a cell came back as a cell object, and `#N/A` in a hidden column reads as blank text. Both are read correctly now.
- **Not every answer was written before the first run.** Committed first (6f3505d): 28 steps, 355 cells. Added after: steps W14a and W14b whole (the lower-is-worse workbook, 36 cells, then more for its chart's floor and top), and readings of the chart's floor and top in W3.1–W3.4, because the first run showed the pictures wrong while every cell read was right. The Glossary count was changed from 33 to 35 when Excel disagreed (the document it came from was wrong), and to 36 when the Status term was added. The run on the Assess redesign added the Periods left out, Status and Scenarios readings. An independent check found this paragraph missing from an earlier version of this document, which said every answer was written first.

What this does not prove

- **A person's hands.** Excel did the arithmetic and drew the charts, but the typing was done through its automation interface. The window (`Thresholds.pyw`) was not walked here; its tests drive it, and three open a real window.
- **Every dial row.** Three dial rows were read against Thresholds (W6–W8); the tests hold every row to `scale.py`.
- **Evidence and Backtest** are written at build by the same Python the expected answers came from: W13 shows they reach Excel intact, not that they are right. The tests and the 12 Oct tie-out cover them.
- **The peer workbook** is not in this walkthrough. It was run end to end and tied to KeyBank's filed Call Report on 12 October (`docs/tie-out/keybank-card-90plus-2026-10-12/`).
- **The bank's data.** Nothing internal was run. The on-site checklist covers that, at the bank.

The steps, as typed

Step	What was typed
W1	Run the build command (section 1), then open the file in Excel.
W2	Read the Thresholds tab. Nothing typed yet.
W3.1	On Assess, pick Credit card in B4. Typed on Assess: B4 Credit card.
W3.2	On Assess, pick Mortgage in B4. Typed on Assess: B4 Mortgage.
W3.3	On Assess, pick Home equity in B4. Typed on Assess: B4 Home equity.
W3.4	On Assess, pick Other consumer in B4. Typed on Assess: B4 Other consumer.
W4	Settings row 6: B6 2007Q4, C6 2012Q4, D6 'financial crisis'. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis.
W5	Also E6 2021Q3, F6 2023Q1, G6 'pandemic forbearance'. Expected to equal LOG.md, 7 Oct: 3.171 / 3.464 / 4.050 / 5.981, and 25 / 22 / 29 / 55 / 4 quarters kept. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis; A43 Credit card; B43 2021Q3; C43 2023Q1; D43 pandemic forbearance.
W6	From the rehearsal settings, change Moderate's half-width 0.5 to 0.75. The dial row for 0.75 on Assess must say the same. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis; A43 Credit card; B43 2021Q3; C43 2023Q1; D43 pandemic forbearance; B58 0.75; C58 1.0; D58 2.5.

W7	From the rehearsal settings, change Moderate-Low step 1.0 to 1.5. The dial row for 1.5 on Assess must say the same. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis; A43 Credit card; B43 2021Q3; C43 2023Q1; D43 pandemic forbearance; B58 0.5; C58 1.5; D58 2.5.
W8	From the rehearsal settings, change High's z 2.5 to 3.0. The dial row for 3 on Assess must say the same. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis; A43 Credit card; B43 2021Q3; C43 2023Q1; D43 pandemic forbearance; B58 0.5; C58 1.0; D58 3.0.
W9a	From the rehearsal settings, H6 0 and K6 'worse'. The median kept (3.757) is a quarter cards actually had, so it sits on two lines and the rule decides its score. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis; A43 Credit card; B43 2021Q3; C43 2023Q1; D43 pandemic forbearance; B58 0; E58 worse.
W9b	From the rehearsal settings, H6 0 and K6 'better'. The median kept (3.757) is a quarter cards actually had, so it sits on two lines and the rule decides its score. Typed on Assess: A42 Credit card; B42 2007Q4; C42 2012Q4; D42 financial crisis; A43 Credit card; B43 2021Q3; C43 2023Q1; D43 pandemic forbearance; B58 0; E58 better.
W10.1	Type Assess!A42 Credit card; Assess!B42 2008Q1; read the Check column and Assess.
W10.2	Type Assess!A43 Credit card; Assess!B43 2021Q1; Assess!C43 2020Q1; Assess!D43 x; read the Check column and Assess.
W10.3	Type Assess!A42 Credit card; Assess!B42 2008Q1; Assess!C42 2012Q4; read the Check column and Assess.
W10.4	Type Assess!B58 -0.1; read the Check column and Assess.
W10.5	Type Assess!D58 0; read the Check column and Assess.
W10.6	Type Assess!C58 one; read the Check column and Assess.
W10.7	Type Assess!A42 Credit card; Assess!B42 1985Q4; Assess!C42 2025Q2; Assess!D42 test; read the Check column and Assess.
W10.8	Type Assess!A44 Mortgage; Assess!B44 1985Q4; Assess!C44 2011Q3; Assess!D44 test; Assess!B4 Mortgage; read the Check column and Assess.
W10.9	Type Assess!C58 0.4; read the Check column and Assess.
W10.10	Type Assess!D58 0.3; read the Check column and Assess.
W10.11	Type Assess!D58 8; read the Check column and Assess.
W10.12	Type Assess!C58 6; read the Check column and Assess.
W11	Read Run C9:C12 and compare with the SHA-256 of each file (certutil -hashfile FILE SHA256).
W12	Chart B3: FALSE. Mortgage's column is #N/A (nothing drawn). Typed on Assess: B3 False.
W13	Read Evidence rows 6-9 (spells) and Backtest row 6.
W14a	Build the score workbook (cards mirrored as 800 - 40 x rate), open it, K6 'worse'. Its lines fall: worst is lowest.
W14b	Build the score workbook (cards mirrored as 800 - 40 x rate), open it, K6 'better'. Its lines fall: worst is lowest.

Every cell read and its expected answer: `docs/walkthrough/EXPECTED.md`. What Excel said: `out/observed.json`, `out/observed-low.json`. To run it again, from `threshold-engine/`:

```
python docs/walkthrough/expected.py
(build both workbooks: the commands are in WALKTHROUGH.md)
powershell -NoProfile -ExecutionPolicy Bypass -File docs/walkthrough/walk.ps1
powershell -NoProfile -ExecutionPolicy Bypass -File docs/walkthrough/walk.ps1 -Workbook docs/walkthroug
h/out/score.xlsx -StepsFile steps-low.json -ObservedFile observed-low.json
```

```
python docs/walkthrough/compare.py
python docs/walkthrough/build_exhibit.py
```